

Methods Used to Measure Hospital Bypass

Most hospital bypass studies aim to identify when patients receive care outside their closest available source of care. However, studies vary substantially in how they define “closest” and how they measure travel beyond local hospitals.

Data Sources

Studies use a variety of data sources including administrative claims, hospital datasets, and geographic datasets.

Data Source	Examples	Purpose
Administrative claims	HCUP SID, Medicare claims, state discharge data	Identify patient admissions and hospital utilization
Hospital datasets	AHA Annual Survey, HCRIS	Identify hospital characteristics and services
Geographic datasets	RUCA, ZCTA, census tracts	Measure patient location and travel

A small number of studies relied primarily on survey data. These studies were unique because they measured patient-reported decisions and the motivation behind the bypass.

Defining Bypass

The literature showed four primary approaches to measuring bypass:

- **Nearest Hospital (most common):** Patients were classified as bypassers if they received care at any hospital other than the geographically nearest hospital.
- **Nearest Capable Hospital (increasingly common):** Recent studies first identify the nearest hospital capable of providing the required service before determining whether bypass occurred. This avoids incorrectly labeling patients whose nearest hospital did not offer the necessary service.
- **Difference-in-Distance or Time Thresholds:** Many recent studies only classify patients as bypassers if they travel substantially farther than necessary. This method reduces misclassification caused by geographic approximation using ZIP code centroids .
- **Hospital Service Area Definitions:** Several studies define bypass as receiving care outside a patient's local Hospital Service Area rather than at an individual hospital.

Common Geographic Measures

Measure	Frequency	Strengths	Limitations
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Straight-line distance	Common in earlier studies	Simple and reproducible	Does not reflect actual travel routes
Driving distance	Common in recent studies	Uses road networks, more realistic	Requires GIS tools
Travel time	Increasingly preferred	Best represents patient burden	Requires more complicated modeling

Most studies estimated patient location using ZIP Code Tabulation Area centroids, although many newer studies used population-weighted centroids or census tracts for more accuracy.

Common Geographic Tools

Frequently used tools and software include:

Tool	Purpose
RUCA Codes	Classify Rurality
ZCTA crosswalk files	Convert ZIP codes to geographic areas
QGIS / ArcGIS	Calculate spatial relationships and travel distances
Google Maps API	Estimate driving distance/time
Geosphere R package	Calculate geographic distance
Stata / SAS	Statistical Analysis

Recent studies show an increasing use of GIS software.

Common Analytical Approach

Most studies followed a similar analytical workflow:

1. Identify patient residence and hospital location.
2. Determine nearest appropriate hospital.
3. Compare the actual hospital used with the nearest hospital.
4. Classify bypass status.
5. Analyze predictors and outcomes associated with bypass.

Common statistical approaches included descriptive analyses of bypass rates, logistic and multivariable regression, propensity score weighting, instrumental variable analyses, and spatial accessibility and network analyses.

Key Takeaways

Overall, methods for measuring hospital bypass have become increasingly sophisticated over time. Research has shifted from simple comparisons of the nearest hospital toward more nuanced approaches that consider hospital capability, meaningful travel differences, and geographic access. GIS advances have improved measurement of patient travel patterns.